IOC Analysis Report

Abstract

This report offers a concise overview of the Emotet malware, a highly adaptive cyber threat. It outlines Emotet's behavior, associated risks, real-world impact, and strategic defense mechanisms. The information is intended to inform cybersecurity strategies in hackathon projects focused on malware detection, incident response, or network hardening.

Malware Type Identification

The IOC *453257c3494addafb39cb6815862403e827947a1e7737eb8168cd10522465deb* is associated with Emotet, a notorious malware family known for acting as a banking Trojan, spam bot, and a distributor of other malware like ransomware (example; Ryuk).

Hash Type

This is a SHA-256 hash (64-character hexadecimal string).

Emotet Overview

First identified in 2014, Emotet began as a banking trojan and has evolved into a modular, polymorphic malware platform capable of delivering additional payloads. It spreads primarily through phishing emails that contain infected attachments (usually Word documents with macros) or malicious links. Once executed, Emotet communicates with remote command-and-control (C2) servers to receive updates and download further malware such as TrickBot or ransomware.

Key Risks and Characteristics

Emotet is particularly dangerous due to its worm-like behavior and ability to evade traditional antivirus solutions through frequent code modifications. It leverages stolen email threads to increase its phishing success rate. Once inside a network, it moves laterally by brute-forcing credentials or exploiting shared drives (CISA, 2021). The malware has been linked to large-scale data theft, spam propagation, financial fraud, and system disruptions. According to the department of justice (2021), Emotet-related incidents can cost organizations up to $1 million to remediate.

Sector-Wide Impact

Emotet has severely affected sectors like healthcare, banking, education, and government. In 2021, international law enforcement disrupted the botnet, citing "hundreds of millions" of dollars in global damages and infections affecting critical infrastructure (U.S. Department of Justice, 2021). Hospitals experienced ransomware outbreaks, while financial firms reported data breaches and fraud.

Protection Strategies

Robust defenses against Emotet require a multi-layered approach. Email filtering and advanced threat protection can block malicious attachments and URLs. Consistent patch

management closes vulnerabilities that Emotet may exploit. Enforcing strong password policies and multi-factor authentication limits lateral movement. Network segmentation ensures that an infected machine cannot spread malware easily to sensitive systems. Offline, encrypted backups are critical for ransomware recovery, and employee training remains one of the most effective long-term defenses (Department of Justice, 2021).

IOC Utilization in Cyber Defense

Indicators of Compromise (IOCs), including file hashes, malicious IPs, and domains, are essential tools in Emotet detection and response. These can be integrated into SIEM systems to trigger alerts or fed into intrusion detection/prevention systems (IDS/IPS) for real-time blocking. Platforms like VirusTotal, MISP, and national cybersecurity advisories regularly update Emotet IOCs. When incidents occur, IOCs help teams locate infections, isolate affected devices, and contain further spread (FireEye, 2020).

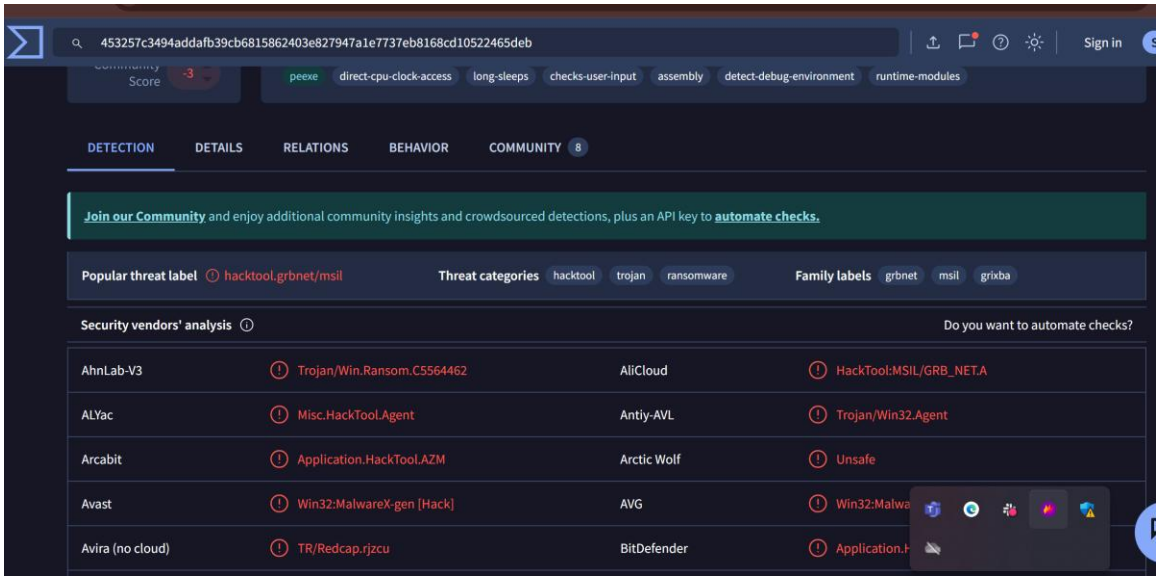
References

- U.S. Department of Justice. (2021, January 27). Emotet botnet disrupted in international cyber operation [Press release].

https://www.justice.gov/opa/pr/emotet-botnet-disrupted-international-cyber-operation

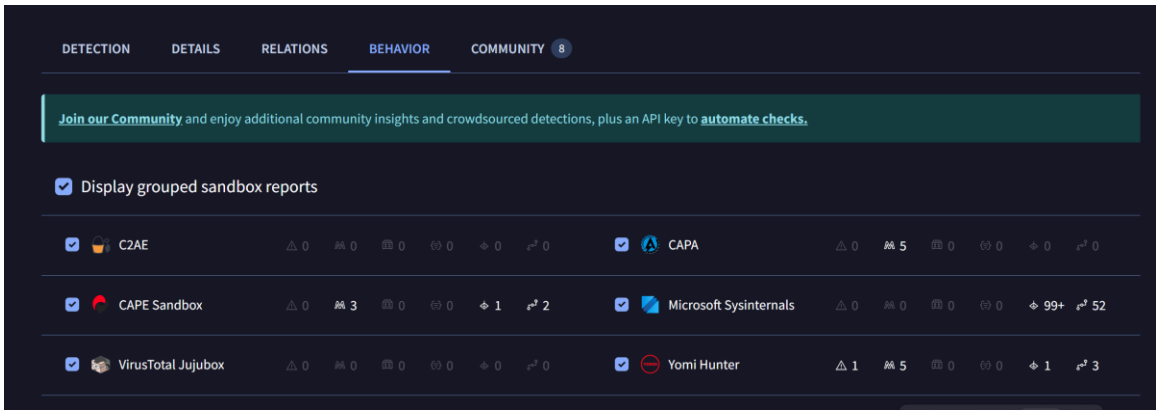
Proof of concept

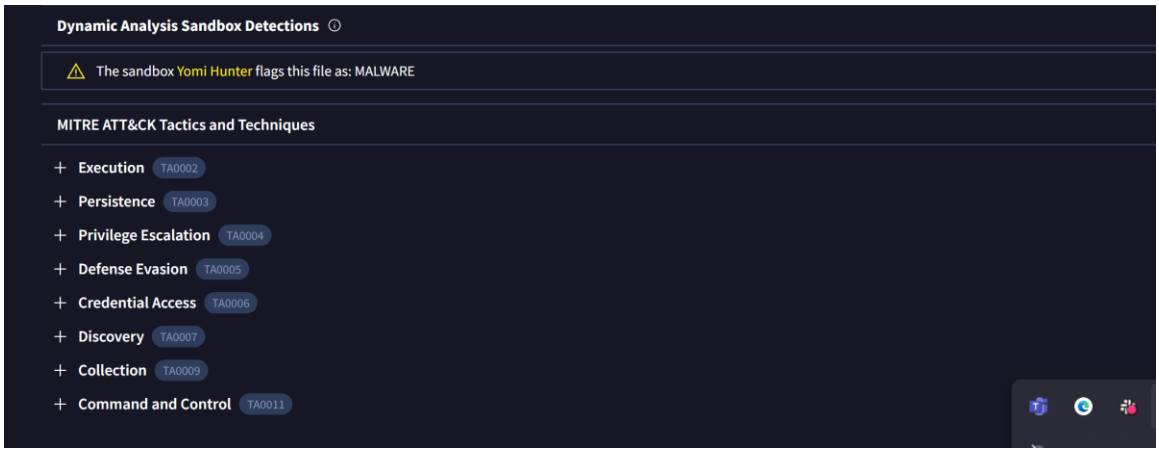
Hash identifier and analysis

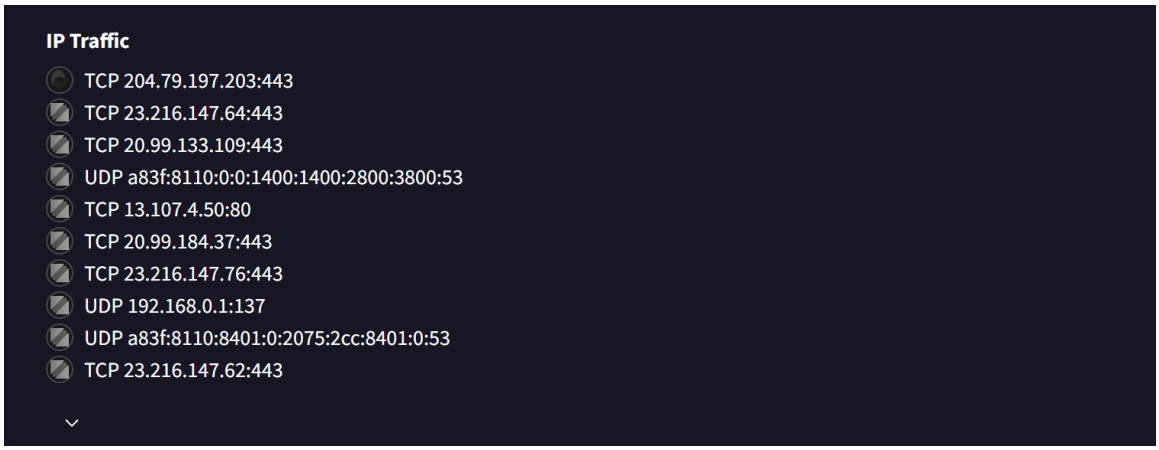


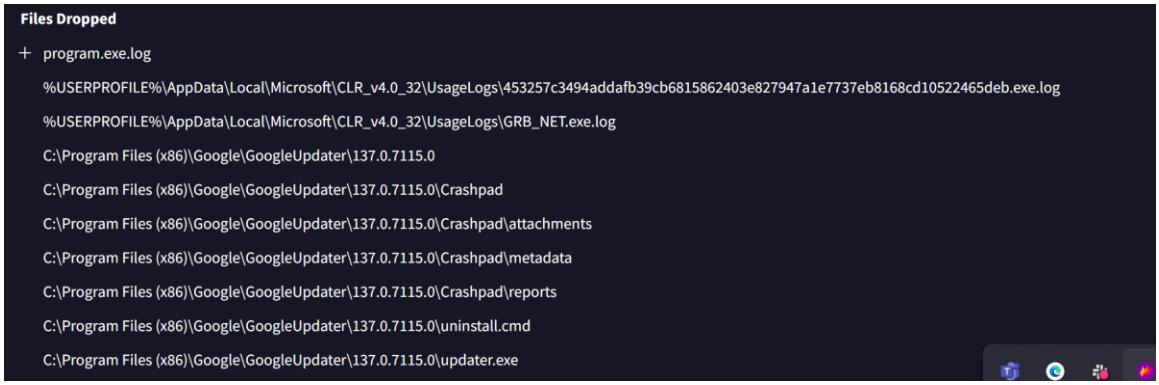
Avira (no cloud)	TR/Redcap.rjzcu	BitDefender	Application.HackTool.AZM
Bkav Pro	W32.AIDetectMalware.CS	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Exe.hacktool.grbnet	DeepInstinct	MALICIOUS
Elastic	Malicious (high Confidence)	Emsisoft	Application.HackTool.AZM (B)
eScan	Application.HackTool.AZM	ESET-NOD32	A Variant Of MSIL/HackTool.GRB_NET.A
Fortinet	W32/Filecoder.PLAY!tr.ransom	GData	Application.HackTool.AZM
Google	Detected	Jiangmin	HackTool.MSIL.bcya
K7AntiVirus	Hacktool (005ab3ac1)	K7GW	Hacktool (005ab3ac1)
Kaspersky	HEUR:HackTool.MSIL.GRBNET.gen	Kingsoft	MSIL.HackTool.GRBNET.gen

Behavior analysis




Dynamic Analysis Sandbox Detections
The sandbox Yomi Hunter flags this file as: MALWARE
MITRE ATT&CK Tactics and Techniques
Execution TA0002
Persistence TA0003
Privilege Escalation TA0004
Defense Evasion TA0005
Credential Access TA0006
Discovery TA0007
Collection TA0009
Command and Control TA0011


IP Traffic
TCP 204.79.197.203:443
TCP 23.216.147.64:443
TCP 20.99.133.109:443
UDP a83f:8110:0:0:1400:1400:2800:3800:53
TCP 13.107.4.50:80
TCP 20.99.184.37:443
TCP 23.216.147.76:443
UDP 192.168.0.1:137
UDP a83f:8110:8401:0:2075:2cc:8401:0:53
TCP 23.216.147.62:443


Files Dropped
program.exe.log
%USERPROFILE%\AppData\Local\Microsoft\CLR_v4.0_32\UsageLogs\453257c3494addafb39cb6815862403e827947a1e7737eb8168cd10522465deb.exe.log
%USERPROFILE%\AppData\Local\Microsoft\CLR_v4.0_32\UsageLogs\GRB_NET.exe.log
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\attachments
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\metadata
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\reports
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\uninstall.cmd
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\updater.exe